General Relativity
Final Examination: Answers

Question 2.

(2 points)

Transformation of the Riemann tensor. Under a change of coordinates $x^\mu \to x'^\mu(x)$ the Riemann tensor, with one upper and three lower indices, transforms as

$$R'^{\mu}{}_{\nu\rho\sigma}(x') = \frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\gamma}{\partial x'^\rho}\frac{\partial x^\delta}{\partial x'^\sigma}\, R^{\alpha}{}_{\beta\gamma\delta}(x)\,. \tag{1}$$

This is a tensor transformation law: one upper index is rotated by $\partial x'/\partial x$ and each lower index by $\partial x/\partial x'$. It holds because the Riemann tensor is built from the connection and its derivatives, and the inhomogeneous terms in the transformation of Γ cancel in the commutator of covariant derivatives.

Ricci tensor. Define $R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$. Apply the transformation law to the rank-(1,3) tensor and contract the upper index $\mu \to \rho$ with the third index ρ (renaming dummy indices as needed):

$$R'_{\nu\sigma}(x') \equiv R'^{\rho}{}_{\nu\rho\sigma} = \frac{\partial x'^\rho}{\partial x^\alpha}\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\gamma}{\partial x'^\rho}\frac{\partial x^\delta}{\partial x'^\sigma}\, R^{\alpha}{}_{\beta\gamma\delta}. \tag{2}$$

The product of the two Jacobians with indices ρ and γ is summed over ρ:

$$\frac{\partial x'^\rho}{\partial x^\alpha}\frac{\partial x^\gamma}{\partial x'^\rho} = \frac{\partial x^\gamma}{\partial x^\alpha} = \delta^\gamma_\alpha, \tag{3}$$

by the chain rule. Therefore

$$R'_{\nu\sigma}(x') = \frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\delta}{\partial x'^\sigma}\,\delta^\gamma_\alpha\, R^{\alpha}{}_{\beta\gamma\delta} = \frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\delta}{\partial x'^\sigma}\, R^{\gamma}{}_{\beta\gamma\delta} \tag{4}$$

$$= \frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\delta}{\partial x'^\sigma}\, R_{\beta\delta}(x). \tag{5}$$

This is exactly the transformation law of a covariant rank-2 tensor, so $R_{\mu\nu}$ is a tensor. □

Question 3.

(3 points)

Convention: $\nabla_\mu V^\rho = \partial_\mu V^\rho + \Gamma_{\mu\nu}{}^\rho V^\nu$, so the first lower index of $\Gamma_{\mu\nu}{}^\rho$ is the derivative index. Write the connection as $\hat\Gamma_{\mu\nu}{}^\rho = \Gamma_{\mu\nu}{}^\rho + K_{\mu\nu}{}^\rho$, with $K_{\mu\nu}{}^\rho = \frac{1}{12} g^{\rho\sigma} H_{\mu\nu\sigma}$.

(a) Torsion. The torsion is $T_{\mu\nu}{}^\rho = \hat\Gamma_{\mu\nu}{}^\rho - \hat\Gamma_{\nu\mu}{}^\rho$. The Levi-Civita part is symmetric, so only K contributes:

$$T_{\mu\nu}{}^\rho = \frac{1}{12} g^{\rho\sigma}\,(H_{\mu\nu\sigma} - H_{\nu\mu\sigma}) = \frac{1}{6} g^{\rho\sigma} H_{\mu\nu\sigma} \neq 0, \tag{6}$$

using $H_{\nu\mu\sigma} = -H_{\mu\nu\sigma}$. Since $H \neq 0$ in general, **the new connection has torsion**.

Metric compatibility. We need $\hat{\nabla}_\mu g_{\nu\lambda} = 0$. Since $\nabla g = 0$ for Levi-Civita,

$$\hat{\nabla}_\mu g_{\nu\lambda} = -K_{\mu\nu}{}^\sigma g_{\sigma\lambda} - K_{\mu\lambda}{}^\sigma g_{\nu\sigma} = -\frac{1}{12}\left(g^{\sigma\rho} H_{\mu\nu\rho}\, g_{\sigma\lambda} + g^{\sigma\rho} H_{\mu\lambda\rho}\, g_{\nu\sigma}\right) \tag{7}$$

$$= -\frac{1}{12}\left(H_{\mu\nu\lambda} + H_{\mu\lambda\nu}\right) = 0, \tag{8}$$

because $H_{\mu\lambda\nu} = -H_{\mu\nu\lambda}$ by total antisymmetry. So **the new connection is metric compatible**, even though it has torsion. (This is the connection with totally antisymmetric torsion used in string theory.)

(b) Geodesics. The geodesic equation for the connection $\hat{\Gamma}$ is

$$\frac{d^2x^\rho}{d\lambda^2} + \hat{\Gamma}_{\mu\nu}{}^\rho \frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0. \tag{9}$$

The extra term is

$$K_{\mu\nu}{}^\rho \frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = \frac{1}{12} g^{\rho\sigma} H_{\mu\nu\sigma} \frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0, \tag{10}$$

because the contraction of an antisymmetric tensor $H_{\mu\nu\sigma}$ with the symmetric product $\dot{x}^\mu \dot{x}^\nu$ vanishes. Hence the geodesic equations coincide with those of Levi-Civita, and the curves (with the same affine parametrisation) are identical. □

Question 4.

(4 points)

(a) The metric functions are singular where $f(r) = 0$ or $r = 0$ (and where f or $1/f$ diverges).

- $r = 2GM$ (the Schwarzschild radius). Here $g_{tt} \to 0$ and $g_{rr} \to \infty$. This is a *coordinate* singularity: it is an artefact of the Schwarzschild (t, r) chart. The curvature invariants are finite there. For example the Kretschmann scalar is

  $$K = R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} = \frac{48G^2M^2}{r^6}, \tag{11}$$

  which is finite at $r = 2GM$. Freely falling observers cross this surface without noticing anything special, so it need not concern you.

- $r = 0$. Here $K \to \infty$, so this is a *genuine curvature singularity* that cannot be removed by a change of coordinates. Inside the horizon r is a time-like coordinate, so any infalling observer is inevitably carried to $r = 0$ in finite proper time. This is the singularity you should be genuinely concerned about.

(b) Start from $v = t + r_*$, with $r_* = r + 2GM \ln\left|\frac{r}{2GM} - 1\right|$. Differentiating,

$$dv = dt + \frac{dr_*}{dr} dr = dt + \frac{dr}{f(r)}, \qquad \text{since} \qquad \frac{dr_*}{dr} = 1 + \frac{2GM}{r - 2GM} = \frac{1}{f}. \tag{12}$$

So $dt = dv - dr/f$. Substituting into the Schwarzschild metric:

$$ds^2 = -f\left(dv - \frac{dr}{f}\right)^2 + \frac{dr^2}{f} + r^2 d\Omega^2 \tag{13}$$

$$= -f\,dv^2 + 2\,dv\,dr - \frac{dr^2}{f} + \frac{dr^2}{f} + r^2 d\Omega^2 \tag{14}$$

$$= -f(r)\,dv^2 + 2\,dv\,dr + r^2\left(d\theta^2 + \sin^2\theta\,d\phi^2\right). \tag{15}$$

This is the ingoing Eddington–Finkelstein form. Its components are regular at $r = 2GM$; the apparent singularity has disappeared. Ingoing null rays are $v = \text{const}$, which cross $r = 2GM$ at finite v, so an infalling observer passes smoothly through the horizon. The only remaining singularity is at $r = 0$, where $K \to \infty$, consistent with part (a). The coordinate r is not a good coordinate at the horizon in the original chart, but v is.

Question 5.

(2 + 1 + 2 points)

(a) The metric is of Friedmann–Lemaître–Robertson–Walker form,

$$ds^2 = -dt^2 + a(t)^2\left(d\chi^2 + \sin^2\chi\,d\Omega^2\right), \qquad a(t) = L\cosh(t/L). \tag{16}$$

The spatial slices are three-spheres of constant positive curvature ($k = +1$), so the geometry is homogeneous and isotropic, as required by the cosmological principle. A cosmological model must satisfy these symmetries, so this is a candidate for our universe. The scale factor is $a(t) = L\cosh(t/L)$. Its minimum radius $a = L$ occurs at $t = 0$, and it expands exponentially at late times, $a \sim \frac{L}{2}e^{|t|/L}$, with $\ddot{a} > 0$. This is accelerated expansion, consistent with the observed late-time universe, which is dominated by a positive cosmological constant. (The model contains no matter, so it is only an approximation to the real universe, in particular to its matter-dominated past.)

(b) Two metrics related by $g_{\mu\nu} = \Omega^2\bar{g}_{\mu\nu}$ with $\Omega^2 > 0$ have the same causal structure. A curve is time-like, null or space-like according to the sign of $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$. Then

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = \Omega^2\,\bar{g}_{\mu\nu}\dot{x}^\mu\dot{x}^\nu, \tag{17}$$

and since $\Omega^2 > 0$ the sign is unchanged. Thus null curves (light cones) are the same in both metrics, so the set of points connected to a given event by causal curves is the same. Conformal transformations change distances and times but not light-cone structure, which is why Penrose diagrams can be drawn in conformally rescaled coordinates.

(c) With $\cosh(t/L) = 1/\cos\eta$, $\eta \in (-\pi/2, \pi/2)$, we have $a = L\cosh(t/L) = L/\cos\eta$. Also $\sinh(t/L) = \sqrt{\cosh^2 - 1} = \tan\eta$ (with the same sign as t). Differentiating $\cosh(t/L) = 1/\cos\eta$:

$$\frac{\sinh(t/L)}{L}\,dt = \frac{\sin\eta}{\cos^2\eta}\,d\eta \quad \Rightarrow \quad \frac{dt}{L} = \frac{\sin\eta}{\cos^2\eta}\cdot\frac{\cos\eta}{\sin\eta}\,d\eta = \frac{d\eta}{\cos\eta}. \tag{18}$$

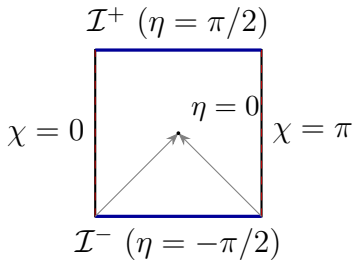
So $dt = L\,d\eta/\cos\eta$, and

$$ds^2 = -\frac{L^2}{\cos^2\eta}d\eta^2 + \frac{L^2}{\cos^2\eta}\left(d\chi^2 + \sin^2\chi\,d\Omega^2\right) \tag{19}$$

$$= \frac{L^2}{\cos^2\eta}\left(-d\eta^2 + d\chi^2 + \sin^2\chi\,d\Omega^2\right). \tag{20}$$

The metric is conformal to the Einstein static universe $\mathbb{R}\times S^3$ metric $-d\eta^2+d\chi^2+\sin^2\chi\,d\Omega^2$, with conformal factor $\Omega = L/\cos\eta$. The range is $-\pi/2 < \eta < \pi/2$, $0 < \chi < \pi$.

The Penrose diagram is the square in the (η,χ) plane, with null rays at 45°:



Each interior point is a 2-sphere of radius $\propto \sin\chi$. The vertical edges $\chi = 0$ and $\chi = \pi$ are the worldlines of antipodal observers, where the spheres shrink to zero. The top and bottom edges are future and past infinity, $\mathcal{I}^{\pm}$. Null rays are diagonal lines, as in any conformal diagram. The diagram shows that de Sitter space has a future infinity but no past singularity.

Question 6.

(1 + 2 + 3 points)

(a) Use the harmonic (Lorenz, de Donder) gauge

$$\partial^\mu \bar{h}_{\mu\nu} = 0. \tag{21}$$

The derivative terms $\partial^\rho\partial_{(\mu}\bar{h}_{\nu)\rho}$ and $\eta_{\mu\nu}\partial^\rho\partial^\sigma\bar{h}_{\rho\sigma}$ then vanish, and the linearised equation reduces to

$$\Box\bar{h}_{\mu\nu} = -16\pi G\,T_{\mu\nu}, \qquad \Box = \eta^{\rho\sigma}\partial_\rho\partial_\sigma = -\partial_t^2 + \nabla^2. \tag{22}$$

In vacuum, $T_{\mu\nu} = 0$, this is a wave equation. Its solutions propagate at the speed of light, $c = 1$. These are gravitational waves. The gauge freedom is used to remove the unphysical components, leaving the transverse-traceless components as the physical waves.

(b) The metric is $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. Test particles that are freely falling stay at fixed coordinate positions x^i in this transverse-traceless gauge, so the only change is the spatial metric. Restricting to t and $z = 0$, the xx and yy components of the metric are

$$g_{xx} = 1 + h_+(t-z)\big|_{z=0} = 1 + h_+(t), \qquad g_{yy} = 1 - h_+(t). \tag{23}$$

The proper distance along the x-axis at fixed time t is

$$\Delta s = \int_0^{L_x}\sqrt{g_{xx}}\,dx = \int_0^{L_x}\sqrt{1+h_+(t)}\,dx = L_x\sqrt{1+h_+(t)}. \tag{24}$$

For $|h_+| \ll 1$, use $\sqrt{1+\epsilon} \approx 1 + \frac{1}{2}\epsilon$:

$$\Delta s \approx L_x \left(1 + \tfrac{1}{2} h_+(t)\right). \tag{25}$$

For the particle at $(0, L_y, 0)$, the integral runs along y, so $g_{yy} = 1 - h_+(t)$ and

$$\Delta s = L_y \sqrt{1 - h_+(t)} \approx L_y \left(1 - \tfrac{1}{2} h_+(t)\right). \tag{26}$$

The two arms change length in opposite senses, which is the quadrupolar "+" pattern of a gravitational wave. The approximation requires $|h_+| \ll 1$ and wavelength $\gg$ the arm lengths, so that h_+ is effectively constant across the apparatus.

(c) *Qualitative description.* A binary of two massive bodies in a circular orbit loses energy and angular momentum by emitting gravitational waves, at a rate given by the quadrupole formula. The orbit therefore shrinks, and the orbital frequency increases.

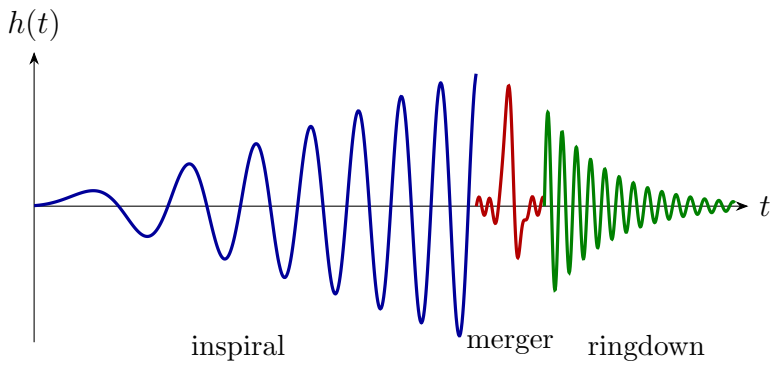
- **Inspiral.** The bodies spiral slowly inward. The orbital frequency and the wave frequency (twice the orbital frequency for quadrupole radiation) increase slowly, and the amplitude grows. This gives a "chirp", a sinusoid with steadily rising frequency and amplitude. The rate of frequency increase depends on the chirp mass of the system.

- **Merger.** When the separation is comparable to the Schwarzschild radii, the bodies plunge together. The motion is highly non-linear and strongly relativistic. The amplitude reaches its maximum, and the frequency reaches its highest value.

- **Ringdown.** The remnant black hole settles down to a stationary Kerr state by emitting quasi-normal modes. The signal is a damped sinusoid whose frequency and damping time are set by the remnant's mass and spin.

Detection. LIGO is a Michelson interferometer with two perpendicular arms. A passing wave stretches one arm and squeezes the other, exactly as computed in (b). The arm length change is $\Delta L/L \sim h/2$, and the two arm lengths change oppositely. The light from the two arms recombines, and the change in the interference fringe is measured. Typical strains are $h \sim 10^{-21}$.

Sketch of the signal. Strain $h(t)$ against time, for an observed binary:



Connection to the physics. The inspiral is the slow decay of the orbit, controlled by the rate of energy loss. The merger is the non-linear coalescence at the peak amplitude. The ringdown is the remnant black hole relaxing, with its quasi-normal frequencies determined by its mass and spin. Measuring the frequency evolution determines the masses of the bodies, and the ringdown determines the properties of the final black hole.